Mahmoud Alyosify

Edge & Efficient AI Engineer

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SUMMARY

Engineer focused on making AI models **fast and cheap enough to run**. Master's thesis cut 47.6% of LLM output tokens and made answers 19–68% faster, with no answer cut off and no retraining. Hands-on with ONNX export, 4-bit fine-tuning (QLoRA), local models on a 4 GB GPU, response caching and C/C++. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30).

SKILLS

Efficiency: inference optimisation, latency and per-token cost measurement, quantisation (4-bit, QLoRA), caching
Runtimes: ONNX, Ollama, FAISS, Hugging Face Transformers, FastAPI, Docker
Machine learning: PyTorch, TensorFlow, scikit-learn, computer vision, LLMs
Systems: C, C++, Python, Linux, Git, AWS, NVIDIA GPUs

PROJECTS

MinimalLM — Master's thesis, published on arXiv — 2026

Python, PyTorch, Hugging Face Transformers, NVIDIA A40

- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).
- Works at answer time, with no retraining: an instruction asks for a complete answer, a stop rule ends answers that run long, and a check lets it stop only at the end of a sentence.
- Cost per token stayed within 0.8% of the original, so the time saved is real; GPU time in fixed batches fell 5–55%. Measured on an NVIDIA A40.

SimCLR-Vision-SSL — self-supervised image model — 2026

PyTorch, SimCLR, SupCon, ONNX, FAISS

- Trained an image model without labels (SimCLR), tested **42 augmentation settings** and reached **84.30% top-1 accuracy** with a semi-supervised extension (SupCon).
- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

Horus-OSINT — fine-tuned LLM on AWS — 2026

Python, Llama-3-8B, QLoRA, PySpark, AWS (EC2, S3, EMR), Terraform

- Fine-tuned Meta-Llama-3-8B with **4-bit QLoRA** on 159,826 records and served it self-hosted on AWS, keeping the model and data in one environment.

RL-Cloud-Autoscaler — deep RL for cloud scaling — 2026

Python, Gymnasium, PPO, DQN

- Built a cloud simulation (Gymnasium) and trained **PPO** and **DQN** agents to balance cost and latency, compared with standard threshold-based autoscaling.

AudioShield — deepfake audio detection

Python, CNN, Streamlit

- Trained a CNN on spectrograms to tell real speech from synthetic speech; served it in a Streamlit app with live predictions and PDF reports.

Also: HORUS Sentinel (multi-agent OSINT) · SCRAP (satellite collision risk, 48+ hours ahead) · Mini-FAT File System (C).

EXPERIENCE

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Ran the assistant on a local **3B model** (Ollama) on a 4 GB GPU; a response cache cut repeat answers from **34.7 s to 1.6 s**.
- Building a RAG assistant that answers employees' questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.

Part-Time Machine Learning Instructor Jul 2025 – Present
National Telecommunication Institute (NTI) Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.

Systems Engineer (Military Service) Feb 2024 – Mar 2025
National Company for Roads Building & Development Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.
- Debugged live transportation software and supported legacy applications to keep downtime to a minimum.

Online Instructor 2020 – Present
Udemy & YouTube (Arabic courses) Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PUBLICATIONS

Say Less, Mean Better: Training-Free Verbosity Control for Language Models
M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026

Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding
M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi); published on arXiv
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS

- Machine Learning Internship, ITIDA / Egypt Makes Electronics (200 hours, 2023)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI & Stanford, 2026)
- IBM AI Developer Professional Certificate (IBM, 2026)
- AWS Certified Cloud Practitioner (Amazon Web Services)
- HCIP-AI training course (Huawei, 80 hours, 2025)

AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)

LANGUAGES

Arabic (native) · English (professional working proficiency)